# Graphing secant, cosecant, and cotangent functions

**ANSWER KEY**

## Core Concepts: Graphing secant, cosecant, and cotangent functions

**1** For the model $f(x) = 6x + 4$ in graphing secant, cosecant, and cotangent functions, evaluate $f(2)$.

**a.** 16 ✓

**b.** 22

**c.** 12

**d.** 20

Substitute $x = 2$: $f(2) = 6(2) + 4 = 16$.

**2** Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

**a.** $x = 1$ and $x = 4$

**b.** $x = 1$ **and** $x = 3$ ✓

**c.** $x = -1$ and $x = -3$

**d.** $x = 4$ and $x = 3$

Factoring $(x - 1)(x - 3) = 0$ yields the roots $x = 1$ and $x = 3$.

**3** A data set recorded for graphing secant, cosecant, and cotangent functions consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

**a.** 15.5

**b.** 16.5

**c.** 17.5 ✓

**d.** 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives $17.5$.

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## Procedural Fluency & Algebraic Manipulation

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**4** For the model $f(x) = 6x + 4$ in graphing secant, cosecant, and cotangent functions, evaluate $f(3)$.

**a.** $22$ ✓

**b.** $28$

**c.** $18$

**d.** $26$

Substitute $x = 3$: $f(3) = 6(3) + 4 = 22$.

**5** Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

**a.** $x = 2$ and $x = 5$

**b.** $x = 2$ **and** $x = 4$ ✓

**c.** $x = -2$ and $x = -4$

**d.** $x = 6$ and $x = 8$

Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

**6** A data set recorded for graphing secant, cosecant, and cotangent functions consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

**a.** $15.5$

**b.** $16.5$

**c.** $17.5$ ✓

**d.** $19.5$

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by $4$ gives $17.5$.

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## Graphical & Geometric Interpretation

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**7** For the model $f(x) = 6x + 4$ in graphing secant, cosecant, and cotangent functions, evaluate $f(4)$.

**a.** $28$ ✓

**b.** $34$

**c.** $24$

**d.** $32$

Substitute $x = 4$: $f(4) = 6(4) + 4 = 28$.

**8** Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

**a.** $x = 3$ and $x = 6$

**b.** $x = 3$ **and** $x = 5$ ✓

**c.** $x = -3$ and $x = -5$

**d.** $x = 8$ and $x = 15$

Factoring $(x - 3)(x - 5) = 0$ yields the roots $x = 3$ and $x = 5$.

**9** A data set recorded for graphing secant, cosecant, and cotangent functions consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

**a.** $15.5$

**b.** $16.5$

**c.** $17.5$ ✓

**d.** $19.5$

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by $4$ gives $17.5$.

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## Exam-Style Problem Solving & Modeling

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**10** For the model $f(x) = 6x + 4$ in graphing secant, cosecant, and cotangent functions, evaluate $f(5)$.

**a.** $34$ ✓

**b.** $40$

**c.** $30$

**d.** $38$

Substitute $x = 5$: $f(5) = 6(5) + 4 = 34$.

**11** Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

**a.** $x = 4$ and $x = 7$

**b.** $x = 4$ **and** $x = 6$ ✓

**c.** $x = -4$ and $x = -6$

**d.** $x = 10$ and $x = 24$

Factoring $(x - 4)(x - 6) = 0$ yields the roots $x = 4$ and $x = 6$.

**12** A data set recorded for graphing secant, cosecant, and cotangent functions consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

**a.** $15.5$

**b.** $16.5$

**c.** $17.5$ ✓

**d.** $19.5$

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by $4$ gives $17.5$.

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